MA1522 Linear Algebra for Computing

Course Overview

Prof Tan Kai Meng

Semester 1, AY2026/27

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It introduces the basic concepts of linear algebra that are routinely applied in computer science, in the context of Euclidean spaces.

The course focuses on the theory behind each algorithm, equipping students to:

- compute with vectors and matrices by hand, and
- use modern computational and AI tools effectively with genuine understanding.

Learning Outcomes

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- reason about subspaces, linear independence, bases, dimension, and the rank of a matrix;
- compute inner products, eigenvalues and eigenvectors, and diagonalise matrices;
- analyse linear transformations between Euclidean spaces and apply these tools to problems in computing.

Teaching Mode

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This differs from the traditional lecture-cum-tutorial format, where students typically attempt tutorial questions as homework before attending tutorials. In Blended Learning, the homework is instead to view the recorded lecture videos before attending 'Lecture'.

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- Slides for each lecture will be uploaded to Canvas, both for review and for those who missed the lecture to catch up.

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- You may attempt them as many times as you like.
- The due dates are only a guide to help you pace yourself — the quizzes remain accessible after they are past due.

Assessments

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- Final Exam: 70%

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- Appeals for missed submissions will **NOT** be handled on an ad-hoc basis; they will be processed together during the Reading Week. (Do **NOT** send me your MC by email.)

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- Each appeal must be supported by appropriate documentation, such as a medical certificate or official confirmation of an approved leave of absence. Appeals without supporting documentation will not be considered.
- Appeals for submissions that fail due to Canvas system errors will **NOT** be granted — always submit early and check that your submission has gone through.

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- It may be typed or handwritten.
- It is submitted in **PDF format**.
- It is submitted successfully to Canvas **before the deadline**.
- Photographs of slides taken during class are **not acceptable**.

Academic Integrity

Do Not Share Submissions

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More severe penalties may also be imposed.

Assessment: Online Quizzes

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- Once a quiz is started, you have **30 minutes** to submit your solutions; after 30 minutes, your solutions will be submitted automatically.

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Each quiz opens at 00:00 and is due at 23:59 the following day:

- Quiz 1 — available Sat 29 Aug, due Sun 30 Aug 2026
- Quiz 2 — available Sat 12 Sep, due Sun 13 Sep 2026
- Quiz 3 — available Sat 3 Oct, due Sun 4 Oct 2026
- Quiz 4 — available Sat 17 Oct, due Sun 18 Oct 2026
- Quiz 5 — available Sat 31 Oct, due Sun 1 Nov 2026

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- It will take place on **24 November 2026, 5:00 pm**.

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- It will take place on **24 November 2026, 5:00 pm**.
- It will cover material from the entire course.
- More details will be announced near the end of the semester.

Lesson Schedule (Weeks 2–6)

This is a tentative schedule for the problems given in each lesson; any changes will be announced in class. You are expected to finish learning the materials on Canvas a lesson ahead of the dates shown below.

Week	Lesson	Date	Day	Sections covered
1	—	—	—	(no lesson; lessons begin in Week 2)
2	1	17 Aug 2026	Mon	Sections 1.1–1.3
	2	19 Aug 2026	Wed	Sections 1.4 & 1.5
3	3	24 Aug 2026	Mon	Sections 2.1–2.3
	4	26 Aug 2026	Wed	Sections 2.4–2.6
4	5	31 Aug 2026	Mon	Sections 2.7–2.9
	6	2 Sep 2026	Wed	Sections 2.8–2.10
5	7	7 Sep 2026	Mon	Sections 3.1–3.3
	8	9 Sep 2026	Wed	Sections 3.3 & 3.4
6	9	14 Sep 2026	Mon	Sections 3.4–3.5
	10	16 Sep 2026	Wed	Sections 3.6 & 3.7
Mid-term break — Sat 19 Sep – Sun 27 Sep 2026 (no class)				

Lesson Schedule (Weeks 7–13)

Week	Lesson	Date	Day	Sections covered
7	11	28 Sep 2026	Mon	Sections 3.7 & 3.8
	12	30 Sep 2026	Wed	Sections 4.1 & 4.2
8	13	5 Oct 2026	Mon	Sections 5.1 & 5.2
	14	7 Oct 2026	Wed	Sections 5.2 & 5.3
9	15	12 Oct 2026	Mon	Sections 5.4 & 5.5
	16	14 Oct 2026	Wed	Section 6.1
10	17	19 Oct 2026	Mon	Sections 6.2 & 6.3
	18	21 Oct 2026	Wed	Sections 6.4 & 6.5
11	19	26 Oct 2026	Mon	Section 7.1
	20	28 Oct 2026	Wed	Sections 7.1 & 7.2
12	21	2 Nov 2026	Mon	Revision: Chapters 1 & 2
	22	4 Nov 2026	Wed	Revision: Chapters 3 & 4
13	—	9 Nov 2026	Mon	*Public Holiday (no lesson)*
	23	11 Nov 2026	Wed	Revision: Chapters 5–6